

KATHMANDU MODEL SECONDARY SCHOOL
FIRST TERM EXAMINATION 2082

Just XI 'Science'
Time: 25 Min
Subject: Com. Mathematics (0071)
SET-B
and the best option among the four options on your answer sheet.

GROUP A: MCQ (1×1=11)

Write the correct option of each question in your answer sheet.

- If $x, y > 0$ and $\frac{1}{x} < \frac{1}{y}$, which one of the following is true?
☒ A) $x > y$ ☐ B) $-\frac{1}{x} > -\frac{1}{y}$ ☐ C) $x < -y$ ☐ D) $x < y$
- Which one of the following is equivalent to $\sim(p \wedge \sim q)$?
☐ A) $\sim(p \wedge q)$ ☐ B) $(p \vee q)$ ☒ C) $(\sim p \vee q)$ ☐ D) $(\sim p \wedge q)$
- What is the value of the $\lim_{x \rightarrow 1} \frac{\sin x}{1-x^2}$?
☐ A) $\frac{\pi}{2}$ ☐ B) π ☐ C) $\frac{2}{\pi}$ ☐ D) Does not exist
- If the function $f(x) = \begin{cases} \frac{\sin^2 ax}{x^2} & \text{for } x \neq 0 \\ 16 & \text{for } x = 0 \end{cases}$ is continuous at $x = 0$, then what is the value of a ?
☐ A) 16 ☒ C) 4 ☐ D) 2 ☐ B) 8
- What is the value of $\lim_{t \rightarrow 0} \frac{\sqrt{3t^2+1}-1}{3t-7}$?
☐ A) $\frac{7}{\sqrt{3}}$ ☒ C) $\frac{1}{\sqrt{3}}$ ☐ D) $\frac{-1}{\sqrt{3}}$ ☐ B) $\sqrt{3}$
- What is the value of $\lim_{y \rightarrow \pi} \frac{\sin y}{2+\cos y}$?
☒ A) 0 ☐ C) π ☐ D) 2 ☐ B) 1
- If $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ and $\mathbb{R}$ have their usual meaning, then which one of the following statements is not true?
☒ A) $\mathbb{Z} \cap \mathbb{Q} = \mathbb{Z}$ ☐ B) $\mathbb{Z} \subset \mathbb{Q}$ ☐ C) $\mathbb{R} \subset \mathbb{Z}$ ☐ D) $\mathbb{Q} \cup \mathbb{R} = \mathbb{R}$

- What is the value of $\lim_{x \rightarrow 0^+} (\frac{1}{x} - \frac{1}{|x|})$?
☒ A) -1 ☐ C) ∞ ☐ D) 1 ☐ B) 0
- What is the value of $\frac{dy}{dx}$, if $y = 2 - 3x^2$?
☐ A) 2 ☒ B) -3 ☐ C) 5 ☐ D) 3
- If $A = \{1, 2, 3\}$, what is the value of $n[P(A)]$?
☐ A) 4 ☐ B) 15 ☐ C) 16 ☒ D) 8
- What does $x \in (B - A)$ mean?
☐ A) $x \in A$ and $x \notin B$ ☐ B) $x \in A$ or $x \notin B$ ☐ C) $x \in B$ and $x \notin A$ ☐ D) $x \notin A$ or $x \in B$

$2^8 \times 2^3 = 2^{11}$



KATHMANDU MODEL SECONDARY SCHOOL

Science

15 Min

Subject: Com. Mathematics (0071)

SET-A

Choose the best option among the four options on your answer sheet.

GROUP 'A' MCQ (1×1=1)

Write the correct option of each question in your answer sheet.

- If $x, y > 0$ and $\frac{1}{x} > \frac{1}{y}$, which one of the following is true?
A) $x > y$ B) $\frac{1}{x} > -\frac{1}{y}$ C) $x < -y$ D) $x < y$
- If $A = \{a, b, c, d\}$, what is the value of $n[P(A)]$?
A) 4 B) 15 C) 16 D) 8
- Which one of the following is equivalent to $\sim(p \vee \sim q)$?
A) $\sim(p \wedge q)$ B) $(p \vee q)$ C) $(\sim p \vee q)$ D) $(\sim p \wedge q)$
- What is the value of the $\lim_{x \rightarrow 1} \frac{\sin x}{1-x}$?
A) 1 B) π C) $\frac{1}{\pi}$ D) Does not exist
- What does $x \in (A - B)$ mean?
A) $x \in A$ and $x \notin B$ B) $x \in A$ or $x \in B$
C) $x \in B$ and $x \notin A$ D) $x \in A$ or $x \in B$
- What is the value of $\lim_{x \rightarrow 0^+} (\frac{1}{x} - \frac{1}{|x|})$?
A) -1 B) 0 C) ∞ D) 1
- If N, Z, Q and R have their usual meaning, then which one of the following statements is not true?
A) $N \cap Q = N$ B) $Z \subset Q$ C) $Q \cup R = R$ D) $R \subset Q$
- What is the value of $\frac{\Delta y}{\Delta x}$, if $y = 2x + 3$?
A) 2 B) 3 C) 5 D) 0

9. If the function $f(x) = \begin{cases} \frac{\sin^2 px}{x^2} & \text{for } x \neq 0 \\ 16 & \text{for } x = 0 \end{cases}$ is continuous at $x = 0$, then what is the value of $\sqrt{p}$?
A) 16 B) 8 C) 4 D) 2

10. What is the value of $\lim_{t \rightarrow \infty} \frac{\sqrt{2t^2+1}}{3t-5}$?
A) $\frac{2}{3}$ B) $\frac{\sqrt{2}}{3}$ C) $\frac{2}{\sqrt{2}}$ D) $\frac{\sqrt{2}}{-3}$

11. What is the value of $\lim_{x \rightarrow \pi} \frac{\sin x}{2 + \cos x}$?
A) 1 B) 0 C) π D) 2

Page XI Science' Subject: Com. Mathematics (20071) P.M.: 75
 time 3 hrs. P.M.: 30
 students are required to give their answers in their own words as far as practicable.
 No figures in the margin indicate full marks.

attempt all the questions.

GROUP 'B' [8x5=40]

12. a) Define symmetric difference of two sets. [1]
 b) For what value of x , $\sqrt{x}$ is real a number? [1]
 c) What are conjugates of conditional $p \Rightarrow q$? [1]
 d) Define a closed interval. [1]
 e) Define the contradiction of a statement. [1]
13. a) If $P \subseteq Q$, then show that $\bar{Q} \subseteq \bar{P}$. [2]
 b) For any three sets X, Y and Z , show that $(X \cup Y) - Z = (X - Z) \cup (Y - Z)$ [2]
14. a) Write the antecedent and the consequent of the statement "If 5 is a natural number then $\frac{1}{5}$ is a rational number." [2]
 b) Let p, q and r be any statements. Show that the following statement is true. $(p \Rightarrow q) \wedge (q \Rightarrow r) \Rightarrow (p \Rightarrow r)$ [3]
15. a) Write any two properties of inequalities. [2]
 b) Solve the inequality: $-3 < 7 - 2x \leq 5$. [2+1]
 Also, represent the solution set on the number line.
16. a) For any two sets A and B , prove that $(A \cap B) = \bar{A} \cup \bar{B}$ [2]
 b) If $A = [-1, 3]$ and $B = (0, 2)$ then find $A \cup B, A \cap B$ and $A - B$ [3]
17. a) Let $h(x)$ be a continuous function. What is the relationship between $\lim_{x \rightarrow p} h(x)$ and $h(p)$? [1]
 b) Define derivative of $f(x)$ with respect to x . [1]
 c) If $\lim_{x \rightarrow a} f(x)$ exists, what is the relationship between $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$? [1]
 d) Define jump discontinuity of a function $f(x)$ at $x = a$. [1]
 e) What is the value of $\lim_{x \rightarrow 0} \frac{e^{\sqrt{x}} - 1}{x}$? [1]

18. a) Evaluate the limit, if it exists. $\lim_{x \rightarrow 16} \frac{\sqrt{x} - 4}{16x - x^2}$ [2]
 b) If $f(x) = \begin{cases} \frac{\sqrt{1+x} - \sqrt{1-x}}{2x+1}, & -1 \leq x < 0 \\ \frac{x}{x-2}, & 0 \leq x \leq 1 \end{cases}$ [2]
 Is continuous at $x = 0$. Find the value of p . [3]

19. a) Find the limit of the following function if exists. [2]
 $f(x) = \begin{cases} 2-x & \text{for } x < 2 \\ 2+x & \text{for } x \geq 2 \end{cases}$ at $x = 2$. [2]

- b) i) Evaluate: $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x}$ [2]
 ii) Is it possible to write $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = \lim_{x \rightarrow 0} x^2 \times \lim_{x \rightarrow 0} \sin \frac{1}{x}$? If not, give reason. [1]

GROUP 'C' [3x8=24]

20. a) For any two statements p and q , prove that $\sim(p \vee q) \equiv \sim p \wedge \sim q$. [2]
 b) Define a Statement. How does it differ from sentence in the context of logic? Explain. [2]
 c) What is the relationship between the existence of limits and the continuity of a function at a particular point? Explain. [4]

21. a) Let $f(x) = \begin{cases} \frac{\tan x - \cot x}{x - \frac{\pi}{4}}, & x \neq \frac{\pi}{4} \\ a, & x = \frac{\pi}{4} \end{cases}$ be continuous at $x = \frac{\pi}{4}$. Find the value of a . [4]

- b) Find the derivative of the function $f(x) = \sqrt[3]{(2-3x)^2}$ by using definition of the derivative. [4]

22. a) Evaluate: $\lim_{x \rightarrow y} \frac{\tan x - \tan y}{x - y}$ [2]
 b) Find from the first principles, the differential coefficient of the function $\cot(\theta - a)$ [2]
 c) If A and B be any two non-empty sets, verify that $A - B$ and $B - A$ are disjoint sets by choosing suitable sets A and B . [3]

Best of Luck

Class: XI 'Science'

Time: 3 hrs.

Subject: Mathematics [0071]

F.M.: 75

P.M.: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP B [8×5=40]

12. a) Define power set with an example.
 b) Define tautology with example.
 c) Under what condition does the inverse of the function exist?
 d) Write $\{x: b > x \geq a\}$ into interval form.
 e) If x is a number, what does $|x|$ geometrically represent? [1+1+1+1+1]
13. a) Show that if $A \subset B$, then $C - B \subset C - A$. [2]
 b) Let p, q, r be any statements. Show that the following statement is true,
 $(p \vee q) \Rightarrow r \equiv (p \Rightarrow r) \wedge (q \Rightarrow r)$ [3]
14. a) For any $x \in \mathbb{R}$ and $p > 0$, prove that
 $-p \leq x \leq p \Rightarrow |x| \leq p$. [2]
 b) Test whether the function
 $f: \mathbb{R} - \{3\} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{2x-1}{x-3}$ is bijective or not. [3]
15. a) Let $g(x) = \frac{x^2+x-6}{|x-2|}$. Find
 i) $\lim_{x \rightarrow 2^+} g(x)$ ii) $\lim_{x \rightarrow 2^-} g(x)$ [2]
 b) Evaluate: $\lim_{t \rightarrow 0} \left(\frac{1}{t\sqrt{1+t}} - \frac{1}{t} \right)$ [3]
16. a) Find from definition the derivative of $\sqrt{\tan(3-5x)}$. [2]
 b) If $(x+t)^5 = x^5 + t^5$, show that $\frac{dx}{dt} = \frac{x}{t}$. [3]
17. a) If $\lim_{x \rightarrow p} f(x)$ and $\lim_{x \rightarrow p} f(x)$ are equal, what can be concluded about
 $\lim_{x \rightarrow p} f(x)$? [3]

- b) Define jump discontinuity of the function $g(x)$ at $x = p$.
- c) What does it mean for the function f to be differentiable at $x = a$?
- d) What does $f'(x) > 0, x \in (a, b)$ geometrically represent?
- e) If $f'(x) = g(x)$, then express $f(x)$ in terms of an antiderivative of $g(x)$.

[1+1+1+1+1]

18. a) Does the function $f(x) = x^5$ have point of inflection? Explain. [2]
 b) Find the local maximum and local minimum values of the function
 $f(x) = x + \frac{25}{x}$. [3]
19. a) If $\int f(x) dx = \sin 2x + \cos^2 x + C$, find $f(x)$. [2]
 b) Evaluate: $\int \tan^5 \theta \sec^7 \theta d\theta$. [3]

GROUP C [3×8=24]

20. a) State the condition for f to be an identity function. [1]
 b) Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 5, 7, 9\}$. Given $f: A \rightarrow B$ is defined by
 $f(x) = 2x + 1$, verify that $f \circ f^{-1} = I_B$ and $f^{-1} \circ f = I_A$ [2]
 where I_X denotes the identity function on set X .
 c) Find the domain and range of $f(x) = \sqrt{x - x^2}$. [4]
21. a) Under what condition is $x = c$ said to be a critical point of the function
 $f(x)$? Also, find the critical points of $f(x) = x^{\frac{2}{3}}(4-x)$. [3]
 b) Solve: $\frac{x^2-3x}{x+2} \leq 0$. Also, represent the solution in number line. [4+1]
22. a) Is it true that a function has infinitely many antiderivatives which differ from one another only by a constant? Illustrate this statement with a suitable example. [2]
 b) What does the Squeeze Theorem say? Evaluate:
 $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x}$ [1+2]
 c) Evaluate: $\int \frac{\sqrt{9-t^2}}{t^2} dt$. [3]

Best of Luck

KATHMANDU MODEL SECONDARY SCHOOL

SECOND TERM EXAMINATION 2082

Class: XI 'Science'

SET 'A'

Time: 25 Min

Subject: Mathematics [0071]

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

GROUP 'A' MCQ [11×1=11]

Re-write the correct option of each question in your answer sheet.

1. Which of the following compound statements is not true?

- A) $\neg(p \wedge q) \equiv (\neg p) \vee (\neg q)$ B) $\neg(p \vee q) \equiv (\neg p) \wedge (\neg q)$
C) $\neg(p \Rightarrow q) \equiv p \vee \neg q$ D) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

2. What is the solution of $|1 - x| < 5$?

- A) $(-\infty, \infty)$ B) $(-4, 6)$
C) $(-\infty, -4) \cup (6, \infty)$ D) $[-4, 6]$

3. For any two nonempty sets A and B, which of the following is true?

- A) $A \cup B \subseteq B$ B) $A \cap B \subseteq A \cup B$
C) $A \subseteq A \cap B$ D) $A \cup \bar{A} = \Phi$

4. Which of the following is the domain of the function $f(x) = \frac{1}{2x-1}$?

- A) $(-\infty, \infty)$ B) $(-\infty, \infty) - \{-\frac{1}{2}\}$
C) $(-\infty, \frac{1}{2}) \cup (\frac{1}{2}, \infty)$ D) $\mathbb{R} - \{1\}$

5. What is the derivative of $e^{\sqrt{\sin x}}$ with respect to x?

- A) $e^{\sqrt{\sin x}}$ B) $\frac{\sin x}{2\sqrt{\cos x}} e^{\sqrt{\sin x}}$
C) $\frac{\cos x}{\sqrt{\sin x}} e^{\sqrt{\sin x}}$ D) $\frac{\cos x}{2\sqrt{\sin x}} e^{\sqrt{\sin x}}$

6. In which interval is the function $f(x) = 1 - \frac{1}{x}$ is increasing?

- A) $(-\infty, \infty)$ B) $(-\infty, \infty) - \{0\}$ C) $(-\infty, 0)$ D) $(0, \infty)$

7. What is the value of $\lim_{x \rightarrow 5} \frac{x^2 - 25}{\sqrt{x} - \sqrt{5}}$?

- A) $10\sqrt{5}$ B) $20\sqrt{5}$ C) $5\sqrt{2}$ D) $-20\sqrt{5}$

8. What is the derivative of $\sin(100 - \pi)$ with respect to x?

- A) $-\cos(\pi - 100)$ B) $\cos(\pi - 100)$
C) $\cos \pi$ D) $\sin \pi$

9. What is the value of the integral $\int \frac{\cos^2(\sqrt{2}x)}{\sin(2\sqrt{2}x)} dx$?

- A) $\frac{1}{2\sqrt{2}} \ln \sec \sqrt{2}x + C$ B) $-\frac{1}{\sqrt{2}} \ln \cos \sqrt{2}x + C$
C) $\frac{1}{2\sqrt{2}} \ln \sin \sqrt{2}x + C$ D) $\frac{1}{2} \ln \sec \sqrt{2}x + C$

10. What is the value of p, if the function $f(x) = \begin{cases} \frac{x}{\ln(1+\sqrt{3}x)} & \text{for } x \neq 0 \\ p & \text{for } x = 0 \end{cases}$ is continuous at $x = 0$?

- A) $\frac{1}{\sqrt{3}}$ B) $\sqrt{3}$ C) 3 D) $\frac{1}{3}$

11. What is value of the integral $\int \sec^2(a - px) dx$?

- A) $\frac{1}{p} \tan(a - px) + C$ B) $-p \tan(a - px) + C$
C) $-\frac{1}{p} \tan(a + px) + C$ D) $-\frac{1}{p} \tan(a - px) + C$

Class: XI 'Science'

Time: 25 Min.

SET 'B'

Subject: Mathematics [0071]

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

GROUP 'A' MCQ [11×1=11]

Re-write the correct option of each question in your answer sheet.

1. What is the value of $\lim_{x \rightarrow 5} \frac{x^2 - 25}{\sqrt{x} - \sqrt{5}}$?
 A) $-20\sqrt{5}$ B) $10\sqrt{5}$ C) $20\sqrt{5}$ D) $5\sqrt{2}$
2. What is the derivative of $\sin(100 - x\pi)$ with respect to x ?
 A) $-\pi \cos(100 - x\pi)$ B) $\pi \cos(100 - x\pi)$
 C) $-\cos(\pi - 100)$ D) $-\pi \cos(\pi - 100)$
3. What is the value of the integral $\int \frac{\cos^2(\sqrt{2}x)}{\sin(2\sqrt{2}x)} dx$?
 A) $-\frac{1}{\sqrt{2}} \ln \cos \sqrt{2}x + C$ B) $\frac{1}{2\sqrt{2}} \ln \sin \sqrt{2}x + C$
 C) $\frac{1}{2} \ln \sec \sqrt{2}x + C$ D) $\frac{1}{2\sqrt{2}} \ln \sec \sqrt{2}x + C$
4. What is the value of p , if the function $f(x) = \begin{cases} \frac{x}{\ln(1+\sqrt{3}x)} & \text{for } x \neq 0 \\ p & \text{for } x = 0 \end{cases}$ is continuous at $x = 0$?
 A) $\frac{1}{3}$ B) $\frac{1}{\sqrt{3}}$ C) $\sqrt{3}$ D) 3
5. What is value of the integral $\int \sec^2(-a + pt) dt$?
 A) $-\frac{1}{p} \tan(a + pt) + C$ B) $\frac{1}{p} \tan(-a + pt) + C$
 C) $\frac{1}{-a} \tan(-a + pt) + C$ D) $-p \tan(a - pt) + C$
6. Which of the following compound statements is not true?
 A) $\sim(p \vee q) \equiv (\sim p) \wedge (\sim q)$ B) $\sim(p \Rightarrow q) \equiv p \vee \sim q$
 C) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ D) $\sim(p \wedge q) \equiv (\sim p) \vee (\sim q)$

7. What is the solution of $|1 - x| < 5$?

A) $[-4, 6]$ B) $(-\infty, \infty)$
 C) $(-4, 6)$ D) $(-\infty, -4) \cup (6, \infty)$

8. For any two nonempty sets A and B , which of the following is true?

A) $A \subseteq A \cap B$ B) $A \cup \bar{A} = \Phi$
 C) $A \cup B \subseteq B$ D) $A \cap B \subseteq A \cup B$

9. Which of the following is the domain of the function $f(x) = \frac{1}{2x-1}$?

A) $(-\infty, \infty) - \{-\frac{1}{2}\}$ B) $(-\infty, \frac{1}{2}) \cup (\frac{1}{2}, \infty)$
 C) $\mathbb{R} - \{1\}$ D) $(-\infty, \infty)$

10. What is the derivative of $e^{\sqrt{\sin x}}$ with respect to x ?

A) $\frac{\cos x}{2\sqrt{\sin x}} e^{\sqrt{\sin x}}$ B) $e^{\sqrt{\sin x}}$
 C) $\frac{\sin x}{2\sqrt{\cos x}} e^{\sqrt{\sin x}}$ D) $\frac{\cos x}{\sqrt{\sin x}} e^{\sqrt{\sin x}}$

11. In which interval is the function $f(x) = 1 - \frac{1}{x}$ is increasing?

A) $(-\infty, 0)$ B) $(0, \infty)$
 C) $(-\infty, \infty)$ D) $(-\infty, \infty) - \{0\}$